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means that the dongle would no longer be required. So what happens is that the BLE mice would directly communicate to the combo radio chip sets inside laptops (for example), thus eliminating the requirement for USB.

How can design engineers incorporate wireless USB solutions into their designs? How will it help them?

If a design engineer wants to connect something to a PC wirelessly, all he needs to do is incorporate a couple of chips at each end. Let us consider he is transferring data through UART and wants to get work out of wireless UART. He could use an MCU to convert UART into a serial peripheral interface (SPI) and then the radio chip will take that SPI data and transmit it to a dongle at the other end, for example, that has an MCU incorporated into it. This MCU will convert the RF into USB, which can be plugged into the PC and straightaway the data will be accessible on the PC. It is very easy to add wireless connectivity to any peripheral because it's not a standard. But note that transfer at high data rates is not feasible. That's the limitation of this technology.

You mentioned that Bluetooth 4.0 would not require a dongle. Why is this not possible in previous versions?

It is possible in previous versions too but Bluetooth 3.0 is much more power-hungry. The throughput of Bluetooth 4.0 is 0.3 Mbps, so it is not designed for data-intensive applications but only for transfer of control signals and low power. The connection efficiency is maximised in 4.0 version. Even in sleep mode, it operates 99 per cent of the time.

What is the most exciting development in USB controllers space that would interest system designers?

The important thing today is USB 3.0, where the speed of data transfer is about 5 Gbps. But we need to apply this to the peripheral device segment in the USB. What I mean by peripheral devices is that if you consider any device like a camera, we can add USB 3.0 connectivity to it. A chip like FX3, which is a very flexible USB 3.0 peripheral controller, will be embedded in the camera (machine-vision camera, for example), which will enable the 'fat' data pipe to be directly connected to the PC transferring large amounts of video data.

For example, on one side we can connect this peripheral controller (FX3) to the PC and on the other side we can connect it to generic programmable interface (GPIS). GPIS can be a parallel interface capable of hooking up to any kind of image sensor. At the end of the day, the data transfer has to come from an image sensor if it is video data.

The image sensor has basically a parallel stream of data. With a few firmware programmable modifications, we can make the FX3 chip connect immediately to the sensor and have data flow from the image sensor to the PC in a matter of minutes, enabling video streaming on the PC seamlessly. ●

(The complete interview can be read at electronicsforu.com)